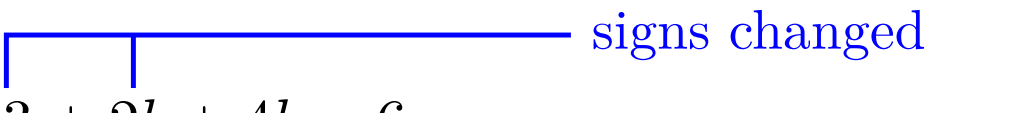$$
\begin{aligned}
-5b-(3-2b)+(4b-6) &= -5b-3+2b+4b-6 \\
&= -5b+2b+4b-3-6 \\
&= b-9
\end{aligned}
$$